

Exercise 9

Given data set $(X_1, Y_1), \dots, (X_n, Y_n) \in \mathcal{X} \times \mathcal{Y}$ where $\mathcal{Y} = \mathbb{R}$, consider split conformal prediction with two given models $\hat{f}_i : \mathcal{X} \rightarrow \mathbb{R}$, the score functions

$$s_i(x, y) = \left| y - \hat{f}_i(x) \right|,$$

and two split conformal prediction sets

$$\mathcal{C}_i(X_{n+1}) = \{y \in \mathcal{Y} \mid s_i(X_{n+1}, y) \leq \hat{q}_i\},$$

where

$$\hat{q}_i = \text{quantile} \left\{ \{s_i(X_j, Y_j)\}_{j=1}^n; (1 - \alpha)(1 + \frac{1}{n}) \right\}.$$

Put

$$L_i = \frac{1}{n} \sum_{j=1}^n (Y_j - \hat{f}_i(X_j))^2$$

and $I = \arg \min_{i \in \{1, 2\}} L_i$. Explain why

$$P \{Y_{n+1} \in \mathcal{C}_I(X_{n+1})\} \geq 1 - \alpha$$

may not hold even if we assume that $(X_1, Y_1), \dots, (X_{n+1}, Y_{n+1})$ are exchangeable.

Solution In one sentence, the reason is that even if we assume that the three random vectors $(X_1, Y_1), \dots, (X_{n+1}, Y_{n+1})$, $s_1(X_1, Y_1), \dots, s_1(X_{n+1}, Y_{n+1})$, and $s_2(X_1, Y_1), \dots, s_2(X_{n+1}, Y_{n+1})$ are all exchangeable, in general it is still not true that $s_I(X_1, Y_1), \dots, s_I(X_{n+1}, Y_{n+1})$ is exchangeable and thus the theorem guaranteeing the marginal coverage does not in general apply.

To show that things can indeed fail, we give a simple counterexample. Put $\mathcal{X} = \mathbb{R}^2$, $n = 1$, $\alpha = \frac{1}{2}$, $X_j = (X_j^{(1)}, X_j^{(2)})$, X_j follows $N_2(0, I_2)$ distribution, $Y_j = X_j^{(1)} + X_j^{(2)}$, $\hat{f}_1(x) = x^{(1)}$ and $\hat{f}_2(x) = x^{(2)}$. It is clear that, for $i = 1, 2$, $\hat{q}_i = s_i(X_1, Y_1)$, $s_i(X_j, Y_j) = \left| X_j^{(2-i)} \right|$, and $L_i = (X_1^{(2-i)})^2$. Thus,

$$\begin{aligned} & P \{s_I(X_2, Y_2) \leq \min(s_1(X_1, Y_1), s_2(X_1, Y_1))\} \\ &= P \{s_1(X_2, Y_2) \leq \min(s_1(X_1, Y_1), s_2(X_1, Y_1)), L_1 < L_2\} \\ &+ P \{s_2(X_2, Y_2) \leq \min(s_1(X_1, Y_1), s_2(X_1, Y_1)), L_1 > L_2\} \\ &= P \left\{ \left| X_2^{(2)} \right| \leq \left| X_1^{(2)} \right| \leq \left| X_1^{(1)} \right| \right\} + P \left\{ \left| X_2^{(1)} \right| \leq \left| X_1^{(1)} \right| \leq \left| X_1^{(2)} \right| \right\} = \frac{1}{6} + \frac{1}{6} = \frac{1}{3} \end{aligned}$$

and

$$P \{Y_2 \in \mathcal{C}_I(X_2)\} = P \{s_I(X_2, Y_2) \leq \min(s_1(X_1, Y_1), s_2(X_1, Y_1))\} = \frac{1}{3} < \frac{1}{2} = 1 - \alpha.$$

This completes the counterexample. ///